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Module 3 Assignment: Arson and Explosives

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Lesson A: Combustion and Arson

1. List three ways that an accelerant used by an arsonist may be detected by a fire investigator (3 marks)

Any three of the following are acceptable for 3 marks

- Unusual quantities of accelerant (especially when located in an unusual area)
- Unusual types of fuel (especially when located in an unusual area)
- Rapid rate of fire growth
- Size of area damaged by fire - large
- distinctive burn patterns in fire debris

2. a Provide examples of a liquid fuel source and a solid fuel source often used by arsonists. (2 marks)

Any one of the following are acceptable

- Liquid fuel source - gasoline, diesel, kerosene, turpentine, butane (1 mark)
- Solid fuel source - wicker, foam, paper, wood (1 mark)

b. Explain why an arsonist would choose each of these fuel sources. (2 marks)

Any one of the following are acceptable

- Liquid fuel source - readily available at stores, easy to use, fast, spreads easily (1 mark)
- Solid fuel source - readily available in homes or buildings, may possibly prevent suspicion, increase rate of fire growth (1 mark)

3. Describe two differences in the behaviour of an adult arsonist and the behaviour of a juvenile arsonist. (4 marks)

Any two of the following are acceptable for 2 marks

Differences - Adult

- Adult arsonists tend to work alone
- Adult arsonist's motive is usually revenge or profit
- Adult arsonist's target is usually buildings

Differences - Juvenile

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- Juvenile arsonists often work in pairs or groups
- Juvenile arsonist's motive is usually vandalism or excitement
- Juvenile arsonist's target is usually schools, abandoned buildings

4. Describe one motive that is characteristic of young arsonists and describe how their behaviour can be stopped. (2 marks)

Any one of the following are acceptable for 2 marks:

- Motives of young arsonists include boredom, curiosity about fire, accidents, peer pressure, or expressions of anger and stress.
- Child arsonist's often will stop their behaviour when counselled by an adult. Recognizing and identifying children who have problems with setting fires deliberately is important because often their behaviour can be stopped through counselling.

Lesson B: Investigating Arson Fires

5. Describe three reasons arson fires are difficult to solve. (3 marks)

Any three of the following answers are acceptable for 3 marks

- Usually done at night
- Committed away from eyewitnesses
- Evidence often destroyed by fire
- Evidence often circumstantial

6. Describe three forms of evidence that indicate that a fire was caused by an arsonist. (3 marks)

Any three of the following answers are acceptable for 3 marks

- Multiple points of origin
- Point of origin near a good supply of oxygen (such as an open window)
- Evidence that fire burned quickly and for a relatively long time
- Empty fuel cans or other evidence of the use of accelerants
- Unusual odours caused by the use of accelerants

"The most common accelerant used in arson fires by far is gasoline because it is easy to obtain and does not cause any suspicions when being purchased."

7. a. What can fire investigators use to identify gasoline residue at a fire scene? (1 mark)

Photo Ionization Detector (1 mark)

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b. Identify a laboratory method that can confirm that evidence from a fire scene contains gasoline (1 mark)

Any one of the following answers are acceptable (1 mark)

- Gas chromatography
- Metal oxide sensor
- Mass spectrometry

8. Why are gas chromatography and mass spectrometry often both used when analyzing unknown evidence from an arson scene? (3 marks)

Most of the accelerant residue found in fire debris involves complex mixtures. (1 mark)
Mass spectrometry can identify only one compound at a time and not a mixture. (1 mark)
However, gas chromatography is very effective at separating the individual compounds in an unknown mixture. (1 mark)

Lesson C. Explosives

9. Describe a fuse, and explain why some explosive devices require one. (2 marks)

A fuse is a length of cord either filled with combustible material or made from combustible material. (1 mark)
Some devices require a fuse to produce heat to cause a reaction between the compounds in the metal casing resulting in an explosion. (1 mark)

10. Identify and describe a major disadvantage of using both the X-ray machine and the metal detector to detect explosives. (2 marks)

Both methods require visual confirmation by a person if a positive reading is made. (1 mark)
Operators of both these devices rely heavily on visual analysis of objects. Thus, explosives that are cleverly hidden may be missed. (1 mark)

11. Could the EGIS system replace the use of X-ray machines or metal detectors in airports? Explain. (3 marks)

No. The EGIS system takes a long period to confirm its findings. This would disrupt airport scheduling.

12. Identify three features of the device used to destroy Pan Am flight 103. (3 marks)

Any three of the following answers are acceptable for 3 marks

- plastic explosive

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- made from Semtex
- hidden in a suitcase
- time-delay ignition/detonation
- about 450 grams in size
- unaccompanied piece of luggage
- hidden inside a radio cassette recorder

Lesson D. Case Study Involving Arson and Explosives

13. Describe how Raymond Lee Oyler was able to set the wildfires without being seriously burned. (3 marks)

- Oyler used a device that was time-delayed. (1 mark)
- It consisted of five to seven paper or wood matches attached around a Marlboro cigarette with duct tape or a rubber band. When the cigarette was lit, it burned slowly until it reached the matches, ignited them, and started a brush fire in about 10 minutes. (2 marks)

14. What have some suggested could have prevented the death of the five firefighters in the Esperanza Fire? (2 marks)

Firefighters should not be sent to protect vacant houses or isolated individual homes. (2 marks)

15. Was the bomb used to destroy the Alfred P. Murrah Federal Building a high explosive or a low explosive, or did it have traits of both? Explain your choice. (3 marks)

- The bomb exhibited traits of both low and high explosives. (1 mark)
- It had traits of a low explosive because it used some easy-to-find materials, especially ammonium nitrate (fertilizer) and nitromethane (fuel). (1 mark)
- It had traits of a high explosive because it was a large bomb, produced powerful shock waves, and used compounds used in mining, especially Tovex and ANFO. (1 mark)

16. Michael Fortier knew about the Oklahoma City Bombing, yet he failed to inform authorities. Explain why he received a light sentence despite this. (3 marks)

Michael Fortier agreed to testify against McVeigh and Nichols in exchange for a modest sentence and immunity for his wife. (3 marks)